Abstract

Objective: To determine how HIV-1 risk groups impact transmitted diversity and the tempo of viral evolution at a population scale.

Methods: We investigated a set of previously described transmission chains (n = 70) using a population genetic approach and tested whether the expected differences in porportions of multi-variant transmissions are reflected by varying proportions of transmitted diversity between men-having-sex-with-men (MSM) and heterosexual (HET) sub-populations, the largest contributors to HIV spread. To assess evolutionary rate differences among different risk groups, we compiled risk group datasets for subtypes A1, B and CRF01_AE and directly compared the absolute substitution rate and its synonymous and non-synonymous components.

Results: There was sufficient demographic signal to inform the transmission model in BEAST using *env* data to compare the transmission bottleneck size between the MSM and HET risk groups. We find no indications for a different proportion of transmitted genetic diversity at the population level between these groups. In the direct rate comparisons between risk groups, however, we consistently recover a higher evolutionary rate in the male dominated risk group compared to the HET datasets.

Conclusions: We find that the risk group composition affects the viral evolutionary rate and therefore potentially also the adaptation rate. In particular, risk group-specific sex ratios, and the variation in within-host evolutionary rates between males and females, impose evolutionary rate differences at the epidemic level, but we cannot exclude a role of varying transmission rates.

Key words

HIV
Risk group
Transmission bottleneck
Evolutionary rate
Genetic diversity

Introduction

The determinants of the human immunodeficiency virus type 1 (HIV-1) evolutionary rate and its variability has been subject to extensive investigation. Because the process that controls which genetic variants survive the genetic bottleneck at transmission determines how the enormous within-host evolutionary potential of HIV-1 is translated into long-term evolution at a population scale level, the impact of transmission dynamics is central to our understanding of HIV evolution. Various observations suggest that transmission has both a drift and selective component [1-18]; the latter was recently highlighted by a rigorous examination of many transmission pairs by Carlson *et al*. [19].

Since transmission is not fully deterministic, differences in the magnitude of the virus population bottleneck at transmission - with multi-variant transmission estimated to be twice as common in MSM (~40%) than in HET contacts (~20%) [2] - may affect the long-term evolution of HIV at the epidemic level. As a case in point, when investigating the within versus between host rate difference for HIV-1, we found a markedly lower rate difference in our subtype C transmission chain (a ~2-fold rate difference) [15] as compared to earlier results based on subtype B data (a ~4 to 5-fold rate difference) [14]. Following the argumentation of Lythgoe and Fraser [13], we hypothesized that the dissimilarity in the magnitude of the rate difference follows from the differences in the underlying biological characteristics associated with transmission between the largely MSM driven subtype B epidemic and predominantly HET driven subtype C epidemic. That is, the smaller the number of viruses being transmitted (in HET), the higher the chance that new infections will be established by variants that avoided the accumulation of mutations in the donors. Consequently, the association between the number of transmitted variants and risk group can bring about risk group composition related differences in the tempo of HIV evolution. This is not limited to the host HLA background but can also involve resistance to antiretroviral therapy combinations (cART) and vaccines.

In addition to transmission dynamics, the factors that influence the overall amount of divergence accumulating between the founder strain(s) and transmitted virus are also likely to impact the among-host evolutionary rate. Within-host evolutionary rates have for example been shown to vary with disease progression [20]. In this respect, it is interesting to note that men tend to have higher set point viral loads (spVL), a predictor of disease progression [21], than women [22-24], a difference that can persist for several years [25]. This suggests a potentially complex interplay between transmission and within-host evolutionary dynamics in determining the tempo of HIV-1 evolution.

Here, we investigate whether the risk group composition can affect the viral evolutionary rate at a population scale. We do this by examining what factor - the transmission dynamics, within-host evolution, or a combination of both - imposes HIV-1 evolutionary rate differences among different risk groups. To explore the impact of the transmission bottleneck, we use a population genetic approach to contrast the loss of genetic diversity at transmission between risk groups. To this end, we compiled a dataset of previously described HIV-1 transmission chains (n = 70, see Table 1). We describe the viral evolutionary histories with a recently introduced transmission model in BEAST [15] and test for transmission bottleneck size differences with a Bayesian hierarchical phylogenetic model (HPM) approach [26] that incorporates fixed effects [27]. We find no support for a difference in the loss of genetic diversity between the HET and MSM groups. To assess the impact of within-host evolution, we compiled risk group-specific datasets of subtypes A1, B and CRF01_AE and tested for differences in substitution rate (Table 2). For subtype B and CRF01_AE we find that HIV evolves slower in HET than in MSM epidemics, and that for subtype A1 the evolutionary rate is also lower in the HET than in the injecting drug users (IDU) sub-epidemics, and this may be associated with the varying proportions of males in the examined datasets. These estimates indicate that within-host evolutionary processes can impact differences in between-host evolutionary rates.

Methods

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Data compilation - transmission chains

To obtain a comprehensive collection of genetic data sets from HIV-1 transmission chains, we employed the search feature of the HIV database (http://www.hiv.lanl.gov/) that allows retrieving intra-host data sets. At the time of the query (November 2012), 953 sets were reported without specific selection criteria. The title and abstract of the published studies were screened and each study that potentially involved clonal sequence data together with known transmission route and/or known infection time frame was subjected to a more detailed screening. We also undertook a literature search based on the references with high potential from the selected studies.

All transmission pairs for which time-stamped clonal sequences of both donor and recipient was available together with at least an upper bound on the transmission interval were grouped under the following risk groups: heterosexual (HET), men having sex with men (MSM) and blood contact (BC). The denominator 'BC' groups together not only those pairs infected through blood transfusion, but also contains transmission pairs involving a bite [28], a knife-fight [29], surgical procedures [30] and malignant injection [31]. A summary overview of the found transmission chains is presented in Table 1 and the transmission chain dataset details are listed in Additional Table 1.

Data compilation - risk group data sets for evolutionary rate estimation

Subtypes have been shown to evolve at different rates [32]. To compare the evolutionary rate between risk groups within subtypes, we downloaded near complete genome datasets for subtype A1, B and CRF01_AE from the Los Alamos HIV database. The other subtypes lacked sufficient data of at least 2 risk groups to allow for a meaningful comparative analysis among risk groups. This discrepancy is particularly present for subtype C where, in contrast to the large amount of HET sequence data, only three MSM near full genomes are available, and where MSM data from smaller genomic fragments such as the *pol* and *env* CDS lack temporal signal (R^2 values from a regression of root-to-tip divergence vs. sampling time are <0.1). The characteristics of the subtype-specific risk group datasets are given in Table 2. We also explored whether patterns of geographical spread may affect the within-subtype evolutionary rate (Supplementary Text 1).

Incorporating the uncertainty of sampling dates and transmission times

In order to apply the same time-scale for all analyses, all sampling and transmission dates were specified in units of days. Often however, sampling dates or transmission intervals are reported only approximately. Arbitrarily choosing for example the midpoint of the potential interval can introduce biases in the bottleneck size estimations because, especially in situations of recipient sampling close to the time of infection, there can be an interaction between the bottleneck size parameter and transmission/divergence times [15]. To avoid this, we made use of the flexibility offered by BEAST [33] to integrate out the transmission and sampling dates constrained by a time interval [34]. This required an extension of the standard approach [34] to accommodate sampling of a single date for a set of taxa representing a single clonal sample. We therefore extended BEAST [33] to handle such situations, and specified uniform priors over a transmission or sampling interval with known boundaries by making use of the Fiebig stage [35], time to seroconversion or symptoms of primary infection (see Supplementary Text 1).

Phylogenetic inference

We used the BEAST software package [33] for all phylogenetic and population genetic inferences. The substitution process was described using a Hasegawa-Kishino-Yano (HKY, [36]) substitution model with discrete Γ-distribution to model among-site rate heterogeneity [37, 38]. We used a recently described transmission model [15] to accommodate the transmission history and to quantify the transmission bottleneck in terms of the loss of genetic diversity at transmission. We adopted a Bayesian hierarchical phylogenetic model (HPM) procedure to share information across transmission chains and specified hierarchical prior distributions on all parameters of the coalescent model. To formally test for differences in bottleneck size among the risk groups we specified a fixed effect on the bottleneck size parameter [27]. This approach only models the evolutionary dynamics throughout the individual transmission chains and does not consider the evolutionary process between

the different data sets. Although we collected data from *gag*, *pol* and *env* from the HET, MSM and BC risk group, we eventually only retained *env* data from the HET and MSM groups because many transmission chain data sets do not capture the early population dynamics (see Supplementary Text 1 for a more details).

The sparse within-host sampling scheme of most transmission chains (Additional Table 1) resulted in poor temporal information to calibrate the molecular clock in many datasets. We therefore follow a slight modification of the approach by Keele *et al*. [1] to arrive at an informative prior distribution on the evolutionary rate on very short time scales. Specifically, we transformed the standard generation time curve [39] to a rate distribution using the experimentally derived reverse transcriptase error rate [40] and set the upper limit to the rate at a generation time of 18 hours [39].

Tutorials describing how to set up the various component of the above analysis can be found at http://beast.bio.ed.ac.uk/ and example XMLs for the various analyses are provided as Additional Material. All XMLs used for the analyses are available from the authors upon request.

Among risk group evolutionary rate comparison

We visually inspected a regression of root-to-tip divergences as a function of sampling time using Path-O-Gen (http://tree.bio.ed.ac.uk/software/pathogen/). This revealed that sufficient temporal signal was present in all but the full genome subtype B datasets. To remove the noise from this dataset, we followed the same procedure as in [15] to arrive at a balanced dataset with clear temporal signal (R^2 for both ≥ 0.48). The evolutionary rate for all datasets was estimated under a relaxed clock model using a lognormal distribution [41] by fitting the HKY model [36] while allowing for gamma-distributed among-site rate heterogeneity [37, 38]. The genome was split into gene-specific partitions to also allow for among-gene rate variation. A Skygrid model was specified as a flexible tree prior [42]. To estimate absolute synonymous and non-synonymous rates for each dataset we applied a renaissance counting procedure [20] following [15] to the non-overlapping parts of the open reading frames (except for nef for which data was missing for many taxa).

Results

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Dataset compilation

We collected data of 70 transmission chains for the most important risk groups involved in HIV spread: the MSM, HET and BC risk groups. The distribution of transmission chains by risk group, subtype and the genomic fragments sequenced are listed in Table 1. The time between infection and sampling of the recipient patients is an important variable for accurately quantifying the loss of diversity at transmission using a population genetic approach [15]. The available number of samples is also important as this determines the amount of information available to the phylogenetic reconstructions. A number of descriptive statistics of these two parameters are summarized in Additional Table 1. Briefly, most recipients were sampled at one time point and both MSM and HET are more frequently sampled earlier since time of infection as compared to BC. Although clonal data was sometimes available for the *gag* and *pol* genomic regions these proved uninformative for estimating the transmission associated loss in genetic diversity (Supplementary Text 1). For this reason, we only report results based on the *env* datasets.

Support for a transmission bottleneck

We first set out to test whether the *env* transmission chain datasets we collected (n = 69) support a bottleneck at transmission. To this purpose, we apply a genealogical transmission chain model that allows the effective viral population size to change upon transmission according to different coalescent models [15]: constant population size (CON, no bottleneck), exponential growth (EXP, the population size upon transmission is an estimable proportion of the donor population size and grows exponentially in the new recipient) and logistic growth (LOG, as in EXP but with logistic growth in the recipient). We independently fit the transmission model with the three different demographic functions to each data set and compare their model fit using marginal likelihood testing (see Supplementary Text 1). This indicates that a model accommodating a bottleneck (EXP or LOG) is supported by 91.3% (63/69) of the *env* transmission chain data sets (Additional Figure 1). There is however a marked difference between the risk groups: whereas we find strong Bayes factor support in 82.35% (28/34) of the HET and 64.3% (18/28) of the MSM transmission events for a model with a bottleneck, this drops to 14.3% (1/7)for the BC risk group. This likely reflects less informative datasets due to longer times between time of infection and time of sampling for the recipients in the BC transmission pairs (see Supplementary Text 1 and Additional Table 1).

Similar amounts of drift and selection in all risk groups

Despite the evidence for a bottleneck in 63/69 (91.3%) of the *env* datasets, unambiguously estimating its size using the transmission model proved difficult in many cases for reasons described in Supplementary Text 1. We therefore restricted our approach to test differences between risk groups to the most informative subset of transmission chains for the *env* region, which were only available for the HET (n = 17) and MSM (n = 11) risk groups (Supplementary Text 1). The demographic function was parametrized according to the best fitting demographic model in the 'best fit' analysis (with the exception described above). To test the robustness of our bottleneck size estimates to demographic model specification, we also performed an analysis consistently applying either an EXP or a LOG function to all datasets in the 'exponential' and 'logistic' analysis respectively. Using our population genetic approach, the magnitude of the bottleneck is estimated as the proportion of the donor effective population size that is transmitted to recipient, but we report the complement of this proportion as the percentage of loss in diversity at transmission.

In our test approach, we allow for sharing of information on the demographic parameters across individuals, but model potential differences in bottleneck sizes among risk groups using a fixed effect. We do not find Bayes factor support for the risk group fixed effect indicating no difference in the loss of genetic diversity in *env* for both risk groups, which is unlikely to be caused by differences in donor viral diversity between both risk groups (Supplementary Text 1). The severity of the bottleneck is estimated 96.8%, 96.8% and 98% for the 'best fit', 'exponential' and 'logistic' analyses respectively, with individual patient estimates ranging from 54% to 99.9%. The average difference in ancestral proportion size estimates between the 'exponential' and 'logistic' analyses was <1%, and there was no trend for either model to consistently estimate higher or lower values for this parameter, indicating that our estimates are robust to the demographic parameterization.

Gender ratio drives risk group evolutionary rate differences

To test for HIV-1 evolutionary rate differences among different risk groups, we collected near full genome data from HET risk groups for subtype A1, B and CRF01_AE, from MSM risk groups for subtype B and CRF01_AE and from IDUs for subtype A1 (Table 2). We consistently find slower HIV evolutionary rates in the HET data sets as compared to MSM. For subtype A1, the HIV evolutionary rate is also slower in the HET risk group compared to the IDU risk group (Figure 1). In order to asses whether the rate differences reflect variation in selective pressure and/or replication rate (generation time), we follow the approach from [20] to obtain posterior estimates of the absolute synonymous (μ_S) and non-synonymous (μ_N) rates for all risk group datasets (see Methods). This reveals both elevated μ_S and μ_N rates in the MSM and IDU datasets when compared to the HET groups for all subtypes, suggesting that the underlying replication rate is lower in HET groups. By comparing the rates with the proportion of males in the datasets, we see a consistently higher rate for a higher proportion of males within each subtype (Figure 2), and find strong Bayes factor support for this (Bayes factors are 8.2, 333.5 and 203.2 for subtypes A1, B and CRF01_AE respectively). There is however no clear linear relationship between rate and proportion of males independent of the subtype (Figure 2 and Table 3), suggesting that other factors confound evolutionary rate differences associated with gender composition at the epidemiological level.

Discussion

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In this study, we set out to investigate whether the population genetic dynamics associated with transmission, within-host viral replication, or a combination of both, are responsible for evolutionary rate differences between risk groups. For the transmission dynamics to impose such differences, we would expect a difference in bottleneck size among the risk groups and we evaluated this based on a collection of HIV-1 transmission chains for the three most important risk groups (HET, MSM and BC) involved in HIV spread. Although XX% *env* transmission chains provide evidence for a transmission-associated population bottleneck, we had to restrict the estimation of its size to the most informative subset for HET and MSM risk groups. Formal testing between the two risk groups does not provide any evidence for differences in loss of diversity at transmission, which is very high in both HET and MSM.

The absence of a bottleneck size difference is in agreement with many founder effects in HET (~80%) and MSM (~60%) transmission [3], a scenario under which many transmissions will generally be represented by a single phylogenetic branch connecting the donor and recipient viral populations. In such circumstances, it is expected the measured loss in genetic diversity at transmission relates to the sampled diversity in the donor and recipient. In this respect, it is important to point out that our comparative analysis included both plasma and PBMC samples. Plasma samples reflect the freely circulating virus, which is usually interpreted as recently generated diversity. PBMCs however may represent an archive of both past and current HIV-1 diversity. These cells can also be co-cultured before DNA-extraction and the resulting *in vitro* picture may not accurately mirror the *in vivo* diversity. Reassuringly however, it has been shown that the different experimental sampling approaches lead to a similar viral genetic composition [43], and there were no specific biases of either plasma or PBMC between our risk group samples. PCR errors or recombination associated with traditional cloning approaches, which can now be avoided by the use of SGA [1], also represent potential sources of error. Other biases may result from differences in the length of infection [44] as well as the therapy history differences in the donor. The lack of support for a difference in the loss of diversity at transmission, however, implies that these potentially confounding factors did not bias the estimates for the risk groups in a particular direction. Also, we measure the bottleneck as the proportion of diversity that is lost at transmission, and in combination with the many founder effects this explains why -despite the larger share of recently infected donors in the MSM sample- we do not find a difference in the size of the bottleneck between the HET and MSM risk groups (Supplementary Text 1). Finally, although the Los Alamos HIV sequence database contains every HIV sequence the world scientific community has ever published and deposited in GenBank, the representativeness of our work depends on what the scientific field has investigated so far.

We estimate the size of the bottleneck in *env* to ~97%. This is in agreement with the findings by Edwards *et al*. [46] who, also relying on a coalescent approach to infer the decrease in genetic diversity at transmission, estimated a loss of >99% of *env* and *gag* diversity in 1 MSM couple. By comparing the estimated diversity at transmission in 9 MSM and 27 mother-to-child (MTC) transmissions they also found that the mode of transmission does not seem to impact the severeness of the transmission associated bottleneck [46]. Taken together with our findings, this indicates that the difference in frequency of multi-variant transmission (~20% between MSM and HET, ~10% between MSM and MTCT [3]) is too limited to set apart the overall transmitted diversity between the risk groups. The comparable amounts of transmitted diversity imply a similar interplay of drift and selection at transmission in the different risk groups and are therefore not expected to lead to evolutionary rate differences at the epidemic level.

The evolutionary rate comparison among risk groups indicates differences between HET and MSM and between HET and IDU, but the direction of these differences may seem counterintuitive in the light of previous findings. There is increasing evidence that the transmission/establishment advantage of ancestral variants leaves its footprint by slowing down the divergence rate among hosts [13-15, 47]. Given that this effect is larger for subtype B than for subtype C [15], we hypothesized that differences in proportion of multi-variant transmission may play an important role. Following this reasoning, we would expect the following overall ranking in the evolutionary rate between risk groups: HET > MSM > IDU. We find however that the evolutionary rate is lower

in HET compared to MSM and also lower in HET compared to IDU, and this is the case for both absolute synonymous and non-synonymous rates (Figure 1). Because synonymous substitution rates - as a marker of viral replication rates - are associated with HIV-1 disease progression and males are predicted to have higher disease progression rates by their viral set-point, we hypothesize that gender ratios may be key to explaining the risk group associated evolutionary rate differences. This implies that within-host divergence rates, together with a general transmission-associated rate slowdown, impact the tempo of evolution at the population level.

Despite higher rates for risk groups with a larger proportion of males within each subtype, this relationship is less clear independent of the subtypes. On the one hand, the individuals sampled in our data sets may not accurately reflect the composition of the risk group population from which the samples were taken. On the other hand, the impact of differences in sex-specific within-host evolution may also be confounded by varying transmission rates by stage of infection. As suggested by Maljkovic-Berry *et al*. [48], in risk groups that generally transmit at an earlier stage of infection, before selective pressures of the immune system have much impact, the rate of HIV-1 evolution on the population level is expected to be lower compared to the rate when transmission usually occurs throughout asymptomatic infection. This provided an explanation for lower HIV-1 subtype A1 rate estimates among IDUs in the former Soviet Union (FSU) as compared to heterosexual transmission in Africa. Although we find opposite differences here between our HET and IDU data set, this does not necessarily contradict their findings because our IDU sampling is fundamentally different and as a consequence does not specifically reflects IDU transmission.

More replication cycles per unit of time in male-dominated epidemics imply more opportunities for non-synonymous mutations to arise and subsequently become fixed under conditions under which they are beneficial. Because the propagation of mutations in a population depends on several factors, however, this may not easily translate into faster adaptation in for example MSM-driven epidemics. One of these factors is the viral genetic background in which mutations are emerging. Also, population genetic theory predicts that the fixation of mutations will mainly be determined by chance events (drift) in small populations whereas in large populations selection will determine the fate of mutations. In this respect, the spread of beneficial mutations will be confounded by population structure, which introduces drift. In the UK, for example, the size of heterosexual transmission networks was found to be smaller than among MSM [49, 50]. Moreover, in a US-based survey it was found that MSM not only have more partners, but also have more concurrent relationships [51]. Just like variation in rates of partner exchange [52], this may impose different levels of drift. The fact that behavioural differences between risk groups are not universal further complicates predictions. In Quebec for example, transmission clustering did not associate with the mode of transmission [53]. It has also been proposed that a high concurrency of partnerships in heterosexuals may explain why HIV has spread so much more extensively in sub-Saharan Africa, than in other regions [54, 55].

In summary, we show that the effect of within-host evolution on the between-host rate dominates over transmission-related events. Because of this, we propose varying gender ratios as driving risk group related evolutionary rate differences. Knowledge of the factors underlying the differences in the evolutionary dynamics of HIV-1 subtypes may help to predict how the virus will evolve in response to large-scale public health interventions such as vaccination or treatment upscaling.

354 *Table 1*

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356 *Title: Overview of the dataset composition for assessment of between risk-group bottleneck size differences.*

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risk group	# transmission chains	gag	pol	env	subtypes
BC	7 (8)	3	-	7	A-F recombinant(1), B(5), C(1)
HET	34 (45)	-	2	34	A1(9), B(5), C(17), F1(1), G(1), unknown(1)
MSM	29 (31)	7	5	28	B(29)
Total	70 (84)	10	7	69	

359

360 The number of transmission chains is listed, with the total number of transmission events between brackets. The
361 majority of the chains (60/70, 85.7%) represent a single transmission event (range: 1 to 6). In some cases, data
362 for multiple genomic regions was available. The number of chains is also given for each genomic region. Most
363 data are available of the *env* region. Sequences were classified with the COMET subtyping tool when no subtype
364 was reported in the original study [45]. The number of chains from each subtype is given between brackets.

365

366 *Table 2*

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368 *Title: Risk group evolutionary rate dataset characteristics.*

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subtype	risk group (# taxa)	date range	% male
A1	HET (44)	1986-2008	29.5
	IDU (23)	1997-2007	55.5
B	HET (54)	1996-2009	44.4
	MSM (73)	1983-2012	100
CRF01_AE	HET (77)	1990-2009	48.7[1]
	MSM (57)	2007-2011	100

371

372 [1] In these datasets there were a number of taxa with unknown gender origins. The reported male
373 proportion does not take these into account.

374 Figure legends
375
376 Figure 1: *Violin plot representation of the risk group evolutionary rates*. HET, MSM and IDU label the estimate
377 of the evolutionary rate for the respective risk group. The means of each rate estimate are indicate by a white
378 circle. Numbers between brackets indicate the fold increase of the mean relative to the HET mean rate estimate.
379 All rates are in units of nucleotide substitutions per site per year * 10^{-3}.
380
381 Figure 2: *Illustration of the effect of the proportion of males on the rate estimate*. The means of the rate
382 estimates are indicated by symbols, and the whiskers delimitate the credible interval. Squares: subtA1; Triangles:
383 CRF01 AE; Circles: subtB. The impact of the male proportion is visualized as the slope between the mean of the
384 rate estimates of the risk groups per subtype. Uncertainty in the estimate of the slope is reflected in the 95%
385 credible interval range between brackets.

Conflict of interest and source of funding

The authors declare that they have no competing interests.
BV was supported by a PhD grant from the Institute for the Promotion of Innovation through Science and Technology in Flanders (IWT Vlaanderen). MAS is partially supported by National Science Foundation grant DMS-1264153. This work was made possible by funding of the Onderzoeksfonds KU Leuven Research Fund KU Leuven. The research leading to these results has received funding from the European Union Seventh Framework Programme [FP7/2007-2013] under Grant Agreement nr. 278433-PREDEMICS and ERC Grant Agreement nr. 260864. This work was supported in part by grants from the Fonds voor Wetenschappelijk Onderzoek Vlaanderen (FWO G.0692.14), by a grant from the Interuniversity Attraction Poles Programme, Belgian State, Belgian Science Policy (IUAP-VI P6/41), by the European Community's Seventh Framework Programme (FP7/2007-2013) under the project 'Collaborative HIV and Anti-HIV Drug Resistance Network' (CHAIN, grant 223131), by KU Leuven (Program Financing no. PF/10/018). Some of the computational resources (Stevin Supercomputer Infrastructure) and services used in this work were provided by Ghent University, the Hercules Foundation and the Flemish Government - department EWI.

Acknowledgements

BV conceived and performed the experiments, drafted and wrote the manuscript. MAS implemented the computational developments. PL participated in the design of the study and helped with the analysis. GB helped with the analyses. PL, GB, KVL and AVD contributed to writing the manuscript. All authors read and approved the final manuscript. We thank Dr. Thomas Leitner for background information on the contents of the Los Alamos HIV sequence database.

List of supplemental digital content

1) Supplemental digital content 1: Supplementary Text 1.docx
2) Supplemental digital content 2: Additional Figure 1.pdf
3) Supplemental digital content 3: Additional Table 1.pdf
4) Supplemental digital content 4: example XML used for model testing (Boeras-Haaland_RW36_con.xml)
5) Supplemental digital content 5: the XML with the logistic growth model specification used for quantifying the bottleneck size using the *env* data (bottleneck_infOnly_mixOnly_env_EXP.xml)
6) Supplemental digital content 6: a standard XML for estimating the evolutionary rate using near full genome data (subtA1_fullGenome_IDU_skygrid_r1.xml)

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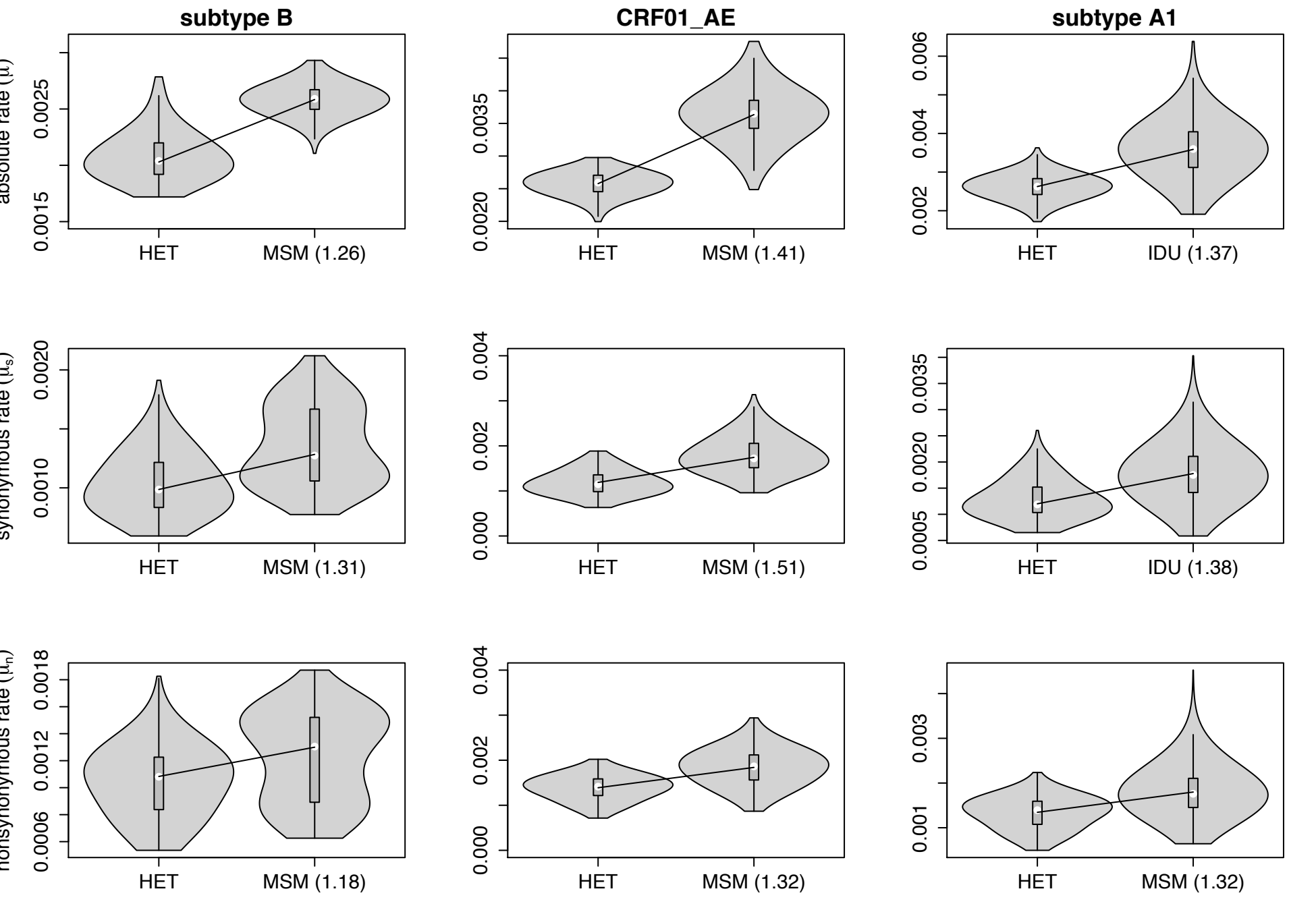
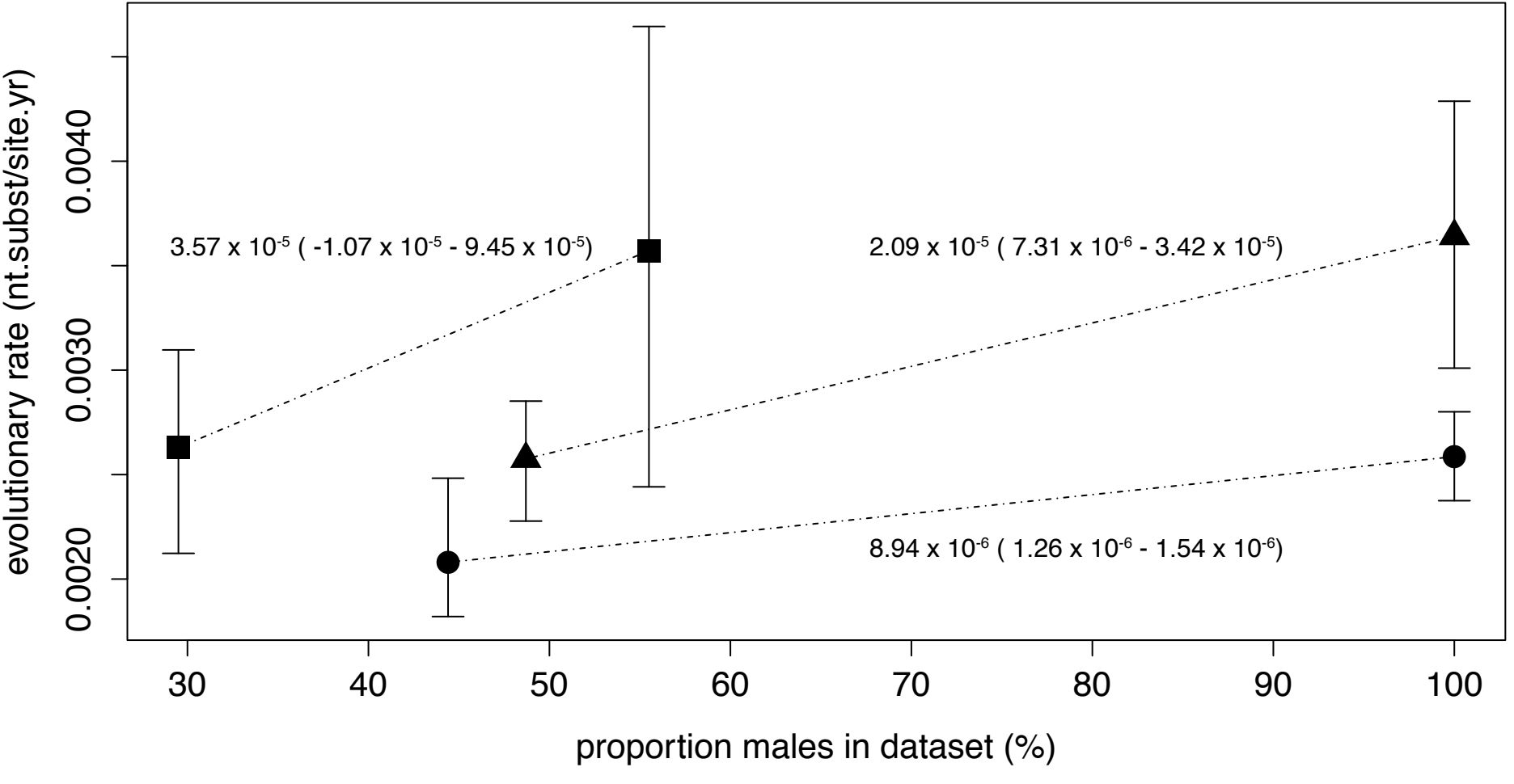

subtype B
CRF01_AE
subtype A1
absolute rate (μ)
synonymous rate (μs)
nonsynonymous rate (μn)
0.0015
0.0025
0.0020
0.0035
0.002
0.004
0.006
0.0010
0.0020
0.000
0.002
0.004
0.0005
0.0020
0.0035
0.0006
0.0012
0.0018
0.000
0.002
0.004
0.001
0.003
HET
MSM (1.26)
MSM (1.41)
IDU (1.37)
MSM (1.31)
MSM (1.51)
IDU (1.38)
MSM (1.18)
MSM (1.32)
MSM (1.32)


evolutionary rate (nt.subst/site.yr)
0.0020
0.0030
0.0040
3.57 x 10-5 (-1.07 x 10-5 - 9.45 x 10-5)
2.09 x 10-5 (7.31 x 10-6 - 3.42 x 10-5)
8.94 x 10-6 (1.26 x 10-6 - 1.54 x 10-6)
30
40
50
60
70
80
90
100
proportion males in dataset (%)